

Basic Concepts Of Python

- 1) Differentiate between break and continue statement with the help of an example.
- 2) Observe the following Python code very carefully and rewrite it after removing all syntactical errors with each correction underlined.

```
DEF execmain():
    x= input("Enter a number:")
    if(abs(x)= x):
        print "You entered a positive number:"
    else:
        x*=-1
        print "Number made positive:"x

execmain()
```

- 3) Write the output of the following Python code:

```
i=5
j=7
x=0
i=i+(j-i)
x=j+i
print x,":",i
j=j**2
x=j+i
i=i+1
print i,":",j
```

- 4) Write the output of the following Python program code:

```
Data =['D','o',' ','I','t',' ','@',' ','1','2','3',' ','!']

for i in range(len(Data)-1):
    if (Data[i].isupper()):
        Data[i]=Data[i].lower()
    elif (Data[i].isspace()):
        Data[i]=Data[i+1]
print Data
```

- 5) Which of the following are valid operators in Python:
(i) ** (ii) */ (iii) like (iv) || (v) is (vi) ^ (vii) between (viii) in
- 6) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
25=Val
for I in the range(0,Val)
    if I%2==0:
        print I+1
    Else:
        print I-1
```

- 7) Find and write the output of the following python code:

```
Text1="AISSCE 2018"
Text2=""
I=0
while I<len(Text1):
    if Text1[I]>="0" and Text1[I]<="9":
        Val = int(Text1[I])
        Val = Val + 1
        Text2=Text2 + str(Val)
    elif Text1[I]>="A" and Text1[I] <="Z":
        Text2=Text2 + (Text1[I+1])
    else:
        Text2=Text2 + "*"
    I=I+1
print Text2
```

- 8) Write the names of any four data types available in Python.
- 9) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
250 = Number
WHILE Number<=1000:
    if Number=>750:
        print Number
    Number=Number+100
    else
        print Number*2
    Number=Number+50
```

- 10) Find and write the output of the following python code:

```
Msg1="WeLcOME"
Msg2="GUeSTs"
Msg3=""
for I in range(0,len(Msg2)+1):
    if Msg1[I]>="A" and Msg1[I]<="M":
        Msg3=Msg3+Msg1[I]
    elif Msg1[I]>="N" and Msg1[I]<="Z":
        Msg3=Msg3+Msg2[I]
    else:
        Msg3=Msg3+"*"
print (Msg3)
```

- 11) Differentiate between Syntax Error and Run-Time Error? Also, write a suitable example in Python to illustrate both.
- 12) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
Val = int(Input("Value:"))
Adder = 0
for C in range(1,Val,3)
    Adder+=C
    if C%2=0:
        Print C*10
    Else:
        print C*
print Adder
```

- 13) Find and write the output of the following python code:

```
Data = ["P",20,"R",10,"S",30]
Times = 0
Alpha = ""
Add = 0
for C in range(1,6,2):
    Times= Times + C
    Alpha= Alpha + Data[C-1]+"$"
    Add = Add + Data[C]
print (Times, Add, Alpha)
```

- 14) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
def Sum(Count) #Method to find sum
    S=0
    for I in Range(1,Count+1):
        S+=I
    RETURN S
print Sum[2] #Function Call
print Sum[5]
```

- 15) Find and write the output of the following python code:

```
for Name in ['John','Garima','Seema','Karan']:
    print (Name)
    if Name[0]== 'S':
        break
    else :
        print ('Completed!')
print ('Welldone!')
```

- 16) What will be the output of the following python code? Explain the try and except used in the code.

```
A=0
B=6
print ('One')
try:
    print ('Two')
    X=B/A
    print ('Three')
except ZeroDivisionError:
    print B*2
    print ('Four')
except:
    print B*3
print ('Five')
```

- 17) Which of the following can be used as valid variable identifier(s) in Python?
(i) 4thSum (ii) Total (iii) Number# (iv) Data
- 18) Name the Python Library modules which need to be imported to invoke the following functions
(i) floor() (ii) randint()
- 19) Rewrite the following code in python after removing all syntax error(s). Underline each correc-

tion done in the code.

```
STRING= "WELCOME"
NOTE=
for S in range[0,8]:
    print STRING(S)
print S+STRING
```

20) Find and write the output of the following python code:

```
TXT = ["20","50","30","40"]
CNT = 3
TOTAL = 0
for C in [7,5,4,6]:
    T = TXT[CNT]
    TOTAL = float (T) + C
    print (TOTAL)
    CNT-=1
```

21) Which of the following can be used as valid variable identifier(s) in Python

(i) total (ii) 7Salute (iii) Que\$tion (iv) global

22) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
TEXT="GREAT
DAY"
for T in range[0,7]:
    print TEXT(T)
print T+TEXT
```

23) List one similarity and one difference between List and Dictionary datatype

24) Rewrite the following Python program after removing all the syntactical errors (if any), underlining each correction.

```
def checkval:
    x = input("Enter a number")
    if x % 2 = 0 :
        print x,"is even"
    else if x<0 :
        print x,"should be positive"
    else ;
        print x,"is odd"
```

25) Out of the following, find those identifiers, which cannot be used for naming Variable or Functions in a Python program:

Total*Tax, While, class, switch, 3rdRow, finally, Column31, _Total

26) Rewrite the following code in python after removing all syntax error(s). Underline each correction done in the code.

```
for Name in [Ramesh, Suraj, Priya]
    IF Name[0]='S':
        print(Name)
```

27) Find and write the output of the following python code:

```
Values=[10,20,30,40]
```

```
for Val in Values:
    for I in range(1, Val%9):
        print(I,"*",end= " " )
    print()
```

- 28) What will be the output of the following python code considering the following set of inputs?
AMAR THREE A123 1200

Also, explain the try and except used in the code.

```
Start=0
while True:
    try:
        Number=int(input("Enter Number"))
        break
    except ValueError:
        Start=Start+2
        print("Reenter an integer")
print(Start)
```

- 29) Find and write the output of the following python code:

```
Numbers=[9,18,27,36]
for Num in Numbers:
    for N in range(1, Num%8):
        print(N,"#",end= " " )
    print()
```

- 30) Rewrite the following code after removing the syntactical errors (if any). Underline each correction.

```
def chksum:
    x= input("Enter a number")
    if (x%2 = 0):
        for i range(2*x):
            print i
        loop else:
            print "#"
```

- 31) State True or False

“Tuple is datatype in Python which contain data in key-value pair.”

- 32) Given the following dictionaries

```
dict_student = {"rno" : "53", "name" : 'Rajveer Singh'}
dict_marks = {"Accts" : 87, "English" : 65}
```

Which statement will merge the contents of both dictionaries?

- (A) dict_student + dict_marks (B) dict_student.add(dict_marks)
(C) dict_student.merge(dict_marks) (D) dict_student.update(dict_marks)

- 33) Consider the given expression:

```
not ((True and False) or True)
```

Which of the following will be correct output if the given expression is evaluated?

- (A) True (B) False (C) NONE (D) NULL

- 34) Select the correct output of the following code:

```
>>> s='mail2kv@kvsangathan.kvs.in'
>>> s=s.split('kv')
>>> op = s[0] + "@kv" + s[2]
>>> print(op)
```

(A) mail2@kvsangathan (B) mail2@sangathan. (C) mail2@kvsangathan. (D) mail2kvsangathan.

35) Which of the following statement(s) would give an error after executing the following code?

```
D={'rno':32, 'name':'Ms Archana', 'subject':['hindi','english','cs'], 'marks':(85,75,89)} #S1
print(D) #S2
D['subject'][2]='IP' #S3
D['marks'][2]=80 #S4
print(D) #S5
```

(A) S1 (B) S3 (C) S4 (D) S3 and S4

36) What will the following expression be evaluated to in Python?

```
print ( round (100.0 / 4 + (3 + 2.55) , 1 ) )
```

(A) 30.0 (B) 30.55 (C) 30.6 (D) 31

37) **str1= "Class" + "Work"**

ASSERTION: Value of str1 will be "ClassWork".

REASONING: Operator '+' adds the operands, if both are numbers & concatenates the string if both operands are strings.

(A) Both A and R are true and R is the correct explanation for A
 (B) Both A and R are true and R is not the correct explanation for A
 (C) A is True but R is False
 (D) A is false but R is True

38) Vivek has written a code to input a number and check whether it is even or odd number. His code is having errors. Rewrite the correct code and underline the corrections made.

```
Def checkNumber(N):
    status = N%2
    return

#main-code
num=int( input(" Enter a number to check :))
k=checkNumber(num)
if k = 0:
    print("This is EVEN number")
else:
    print("This is ODD number")
```

39) Given is a Python string declaration:

```
message='FirstPreBoardExam@2022-23'
```

Write the output of: `print(message[ : : -3].upper())`

40) Write the output of the code given below:

```
d1={'rno':25, 'name':'dipanshu'}
d2={'name':'himanshu', 'age':30,'dept':'mechanical'}
d2.update(d1)
```

```
print(d2.keys())
```

41) Predict the output of the Python code given below:

```
data=["L",20,"M",40,"N",60]
times=0
alpha=""
add=0
for c in range(1,6,2):
    times = times + c
    alpha = alpha + data [c-1] + "@"
    add = add + data[c]
print (times, add, alpha)
```

42) Predict the output of the Python code given below:

```
L=[1,2,3,4,5]
Lst=[]
for i in range(len(L)):
    if i%2==1:
        t=(L[i],L[i]**2)
        Lst.append(t)
print(Lst)
```

43) Predict the output of the code given below:

```
s="C++VsPy"
m=""
for i in range(0, len(s)):
    if (s[i] >= 'a' and s[i] <= 'm'):
        m = m +s[i].upper()
    elif (s[i] >= 'n' and s[i] <= 'z'):
        m = m +s[i-1]
    elif (s[i].isupper()):
        m = m + s[i].lower()
    else:
        m = m +'&'
print(m)
```

44) State True or False

“Python has a set of keywords that can also be used to declare variables”

45) Which of the following is not a valid python operator?

a) % b) in c) # d) **

46) What will be the output of the following python dictionary operation?

```
data = {'A':2000, 'B':2500, 'C':3000, 'A':4000}
print(data)
```

a) {'A':2000, 'B':2500, 'C':3000, 'A':4000} b) {'A':2000, 'B':2500, 'C':3000}
c) {'A':4000, 'B':2500, 'C':3000} d) It will generate an error.

47) print(True or not True and False)

Choose one option from the following that will be the correct output after executing the above python expression.

a) False b) True c) or d) not

48) Select the correct output of the following python code:

```
str="My program is program for you"
t = str.partition("program")
print(t)
```

- a) ('My ', 'program', ' is ', 'program', ' for you') b) ('My ', 'program', ' is program for you')
 c) ('My ', ' is program for you') d) ('My ', ' is ', ' for you')

49) Which of the following operations on a string will generate an error?

- a) "PYTHON"*2 b) "PYTHON" + "10" c) "PYTHON" + 10 d) "PYTHON" + "PYTHON"

50) What will be the output of the following python expression?

```
print(2**3**2)
```

- a) 64 b) 256 c) 512 d) 32

51) Assertion (A): A variable declared as global inside a function is visible with changes made to it outside the function.

Reasoning (R): All variables declared outside are not visible inside a function till they are redeclared with global keyword.

- (A) Both A and R are true and R is the correct explanation for A
 (B) Both A and R are true and R is not the correct explanation for A
 (C) A is True but R is False
 (D) A is false but R is True

52) Sameer has written a python function to compute the reverse of a number. He has however committed a few errors in his code. Rewrite the code after removing errors also underline the corrections made.

```
def reverse(num):
    rev = 0
    While num > 0:
        rem == num %10
        rev = rev*10 + rem
        num = num//10
    return rev
print(reverse(1234))
```

53) What will be the output of the following string operation.

```
str="PYTHON@LANGUAGE"
print(str[2:12:2])
```

54) Write the output of the following code.

```
data = [1,2,4,5]
for x in data:
    x = x + 10
print(data)
```

55) Predict the output of the following python code:

```
data = [2,4,2,1,2,1,3,3,4,4]
d = {}
for x in data:
    if x in d:
        d[x]=d[x]+1
    else:
        d[x]=1
print(d)
```


56) What will be the output of the following python program?

```
str = ""
name = "9@Days"
for x in name:
    if x in "aeiou":
        str+=x.upper()
    elif not x.isalnum():
        str+="*"
    elif x.isdigit():
        pass
    else:
        str+=x.lower()
print(str)
```

57) What will the following code do?

```
dict={"Exam":"AISSCE", "Year":2022}
dict.update({"Year":2023} )
```

- a. It will create new dictionary dict={"Year":2023} and old dictionary will be deleted
- b. It will throw an error and dictionary cannot be updated
- c. It will make the content of dictionary as dict={"Exam": "AISSCE", "Year":2023}
- d. It will throw an error and dictionary and do nothing

58) What will be the value of the expression: 14+13%15

59) Select the correct output of the code:

```
a= "Year 2022 at All the best"
a = a.split('2')
a = a[0] + ". " + a[1] + ". " + a[3]
print (b)
```

- (a) Year . 0. at All the best (b) Year 0. at All the best
- (c) Year . 022. at All the best (d) Year . 0. at all the best

60) Which of the following statement(s) would give an error after executing the following code?

```
S="Welcome to class XII" # Statement 1
print(S) # Statement 2
S="Thank you" # Statement 3
S[0]= '@' # Statement 4
S=S+"Thank you" # Statement 5
```

- (a) Statement 3 (b) Statement 4 (c) Statement 5 (d) Statement 4 and 5

61) What will be the output of the following expression?

24//6%3 , 24//4//2 , 48//3//4

- a) (1,3,4) b) (0,3,4) c) (1,12,Error) d) (1,3,#error)

62) Write the output of following code and explain the difference between a*3 and (a,a,a)

```
a=(1,2,3)
print(a*3)
print(a,a,a)
```

63) What will be the output of following program:

```
s="welcome2kv"
n = len(s)
m=""
for i in range(0, n):
    if (s[i] >= 'a' and s[i] <= 'm'):
        m = m +s[i].upper()
    elif (s[i] >= 'n' and s[i] <= 'z'):
        m = m +s[i-1]
    elif (s[i].isupper()):
        m=m+s[i].lower()
    else:
        m=m+ '#'
print(m)
```

64) State True or False: “A dictionary key must be of a data type that is mutable.”

65) What will be the datatype of d, if `d = (15)` ?

(a) int (b) tuple (c) list (d) string

66) Consider the given expression:

```
not 5 or 4 and 10 and 'bye'
```

Which of the following will be correct output if the given expression is evaluated?

(a) True (b) False (c) 10 (d) 'bye'

67) Which of the following statement(s) would give an error after executing the following code?

```
d = {"A" : 1, "B": 2, "C": 3, "D":4} #statement 1
sum_keys = 0 #statement 2
for val in d.keys(): #statement 3
    sum_keys = sum_keys + val #statement 4
print(sum_keys)
```

(a) Statement 1 (b) Statement 2 (c) Statement 3 (d) Statement 4

68) Rewrite the following code in python after removing all the syntax errors. Underline each correction done in the code.

```
num1, num2 = 10
While num1 % num2 = 0
    num1+= 20
    num2+= 30
Else:
    print('hello')
```

69) Given is a Python List declaration:

```
lst1= [39, 45, 23, 15, 25, 60]
```

What will be the output of: `print(lst1.index(23))` ?

70) Write the output of the code given below:

```
x = ["rahul", 5, "B", 20, 30]
x.insert( 1, 3)
x.insert( 3, "akon")
print(x[2])
```

71) State True or False “Python language is Cross platform language.”

72) Which of the following is an invalid identifier in Python?

- (a) Max_marks (b) Max-marks (c) Maxmarks (d) _Max_Marks

73) Predict the output.

```
marks = {"Ashok":98.6, "Ramesh":95.5}
print(list(marks.keys()))
```

- (a) 'Ashok', 'Ramesh' (b) 98.6, 95.5 (c) ['Ashok','Ramesh'] (d) ('Ashok','Ramesh')

74) Consider the given expression:

```
not True and False or not True
```

Which of the following will be correct output if the given expression is evaluated?

- (a) True (b) False (c) NONE (d) NULL

75) Write the output:

```
myTuple = ("John", "Peter", "Vicky")
x = "#".join(myTuple)
print(x)
```

- (a) #John#Peter#Vicky (b) John#Peter#Vicky (c) John#Peter#Vicky# (d) #John#Peter#Vicky#

76) Which of the following statement(s) would give an error after executing the following code?

```
S="Welcome to my python program" # Statement 1
print(S) # Statement 2
S="Python is Object Oriented programming" # Statement 3
S= S * "5" # Statement 4
S=S+"Thank you" # Statement 5
```

- (a) Statement 3 (b) Statement 4 (c) Statement 5 (d) Statement 4 and 5

77) What will the following expression be evaluated to in Python?

```
print(2**3 + (5 + 6)**(1 + 1))
```

- (a) 129 (b) 8 (c) 121 (d) None

78) Given is a Python string declaration:

```
str="CBSE Examination@2022"
```

Write the output of: `print(str[-1:-15:-2])`

79) Write the output of the code given below:

```
d = {"name": "Akash", "age": 16}
d['age'] = 27
d['city'] = "New Delhi"
print(d.items())
```

80) Given the following dictionaries

```
dict_fruit={"Kiwi":"Brown", "Cherry":"Red"}
dict_vegetable={"Tomato":"Red", "Brinjal":"Purple"}
```

Which statement will merge the contents of both dictionaries?

- (a) dict_fruit.update(dict_vegetable) (b) dict_fruit + dict_vegetable
(c) dict_fruit.add(dict_vegetable) (d) dict_fruit.merge(dict_vegetable)

81) State True or False:

“In a Python program, if a break statement is given in a nested loop, it terminates the execution of all loops in one go.”

82) What will be the output of the following statement:

```
print(3-2**2**3+99/11)
```

- a. 244 b. 244.0 c. -244.0 d. Error

83) Select the correct output of the code:

```
s = "Python is fun"
l = s.split()
s_new = "-".join([l[0].upper(), l[1], l[2].capitalize()])
print(s_new)
```

Options:

- a. PYTHON-IS-Fun b. PYTHON-is-Fun c. Python-is-fun d. PYTHON-Is -Fun

84) Which of the following will delete key-value pair for key = “Red” from a dictionary D1?

- a. delete D1(“Red”) b. del D1[“Red”] c. del.D1[“Red”] d. D1.del[“Red”]

85) Consider the statements given below and then choose the correct output from the given options:

```
pride="#G20 Presidency"
print(pride[-2:2:-2])
```

Options

- a. ndsr b. ceieP0 c. ceieP d. yndsr

86) Which of the following statement(s) would give an error during execution of the following code?

```
tup = (20,30,40,50,80,79)
print(tup) #Statement 1
print(tup[3]+50) #Statement 2
print(max(tup)) #Statement 3
tup[4]=80 #Statement 4
```

Options:

- a. Statement 1 b. Statement 2 c. Statement 3 d. Statement 4

87) What possible outputs(s) will be obtained when the following code is executed?

```
import random
myNumber=random.randint(0,3)
COLOR=["YELLOW","WHITE","BLACK","RED"]
for I in COLOR:
    for J in range(1,myNumber):
        print(I,end="*")
    print()
```

Options:

- a. RED* / WHITE* / BLACK* / RED* b. YELLOW* / WHITE* / BLACK* / RED*
c. WHITE* WHITE* / YELLOW* / YELLOW* / BLACK* BLACK* / RED* RED*

d. YELLOW* / WHITE*WHITE* / BLACK* BLACK* BLACK* / RED* RED* RED* RED* RED*

88) Consider the code given below:

```
b=100
def test(a):
    ----- # missing statement
    b=b+a
    print(a,b)
test(10)
print(b)
```

Which of the following statements should be given in the blank for #Missing Statement, if the output produced is 110?

Options:

a. global a b. global b=100 c. global b d. global a=100

89) Q17 and 18 are ASSERTION AND REASONING based questions. Mark the correct choice as

- (a) Both A and R are true and R is the correct explanation for A
- (b) Both A and R are true and R is not the correct explanation for A
- (c) A is True but R is False
- (d) A is false but R is True

Assertion(A): List is an immutable data type

Reasoning(R): When an attempt is made to update the value of an immutable variable, the old variable is destroyed and a new variable is created by the same name in memory.

Assertion(A): Python standard library consists of number of modules.

Reasoning(R): A function in a module is used to simplify the code and avoids repetition.

90) The code given below accepts a number as an argument and returns the reverse number. Observe the following code carefully and rewrite it after removing all syntax and logical errors. Underline all the corrections made.

```
define revNumber(num):
    rev = 0
    rem = 0
    While num > 0:
        rem ==num %10
        rev = rev*10 + rem
        num = num//10
        return rev
print(revNumber(1234))
```

91) Predict the output of the following code:

```
S = "LOST"
L = [10,21,33,4]
D={}
for I in range(len(S)):
    if I%2==0:
        D[L.pop()] = S[I]
    else:
        D[L.pop()] = I+3

for K,V in D.items():
    print(K,V,sep="*")
```

92) Write the Python statement for each of the following tasks using BUILT-IN functions/methods only:

(i) To insert an element 200 at the third position, in the list L1.

(ii) To check whether a string named, message ends with a full stop / period or not.

93) Predict the output of the following code:

```
def Changer(P,Q=10):
    P=P/Q
    Q=P%Q
    return P

A=200
B=20
A=Changer(A,B)
print(A,B, sep='$')
B=Changer(B)
print(A,B, sep='$', end='###')
```

94) Predict the output of the Python code given below:

```
Text1="IND-23"
Text2=""
I=0
while I<len(Text1):
    if Text1[I]>="0" and Text1[I]<="9":
        Val = int(Text1[I])
        Val = Val + 1
        Text2=Text2 + str(Val)
    elif Text1[I]>="A" and Text1[I]<="Z":
        Text2=Text2 + (Text1[I+1])
    else:
        Text2=Text2 + "*"
    I+=1
print(Text2)
```